Abstract

Stroke survivors and individuals with Parkinson's disease (PD) commonly experience gait impairments that reduce mobility, independence, and quality of life. Although Rhythmic Auditory Stimulation (RAS) improves gait characteristics such as cadence, symmetry, and walking speed, few wearable systems integrate Artificial Intelligence (AI) to adapt RAS in real time. This study developed a low-cost AI-driven RAS system using three Micro:bits. Two ankle-mounted devices use machine learning (ML) to classify walking speed into one of eleven gait categories, while a third Micro:bit processes bilateral gait data to generate adaptive metronomic cadence outputs and calculate a Symmetry Index (SI) from left and right step intervals. RAS is activated when gait asymmetry exceeds a clinically established threshold. Preliminary testing showed that the AI model distinguished multiple walking speed conditions and generated cadence outputs that increased linearly with walking speed. Simulated asymmetric gait trials produced elevated SI values and successfully triggered RAS activation. These results demonstrate the feasibility of combining AI and RAS in an affordable wearable system for adaptive gait assistance. Future work will further validate the system and evaluate its effectiveness in neurorehabilitation.

Introduction

While extensive research on the implementation of music therapy in neurorehabilitation and the integration of Artificial Intelligence in music therapy is administered, little is known surrounding the role of AI in Rhythmic Auditory Stimulation (RAS). With fifteen million annual

cases of stroke and an estimated $100 billion spent yearly for treatment (Zhao et al., 2025) as well as approximately one third of all stroke survivors left with permanent disabilities (Woo-ko, 2016), methods for neural rehabilitation are highly relevant. Additionally, stroke survivors and patients with Parkinson's disease face considerable disparities regarding quality of life (Kitatani et al., 2022) and lose independence due to issues with walking speed, symmetry and coordination issues, stride-to-stride variation issues, and more (Pacilli et al., 2025).

Furthermore, socioeconomic burdens are almost always unavoidable for these patients (Pacilli et al., 2025; Liang, 2025; Zhao et al., 2025) due to healthcare costs and frequent inability to work, presenting the additional problem of healthcare accessibility. Therefore, it is conclusive that financial accessibility of rehabilitation for stroke survivors and Parkinson's disease patients is drastically relevant, and while music therapy and Rhythmic Auditory Stimulation is shown to be incredibly beneficial for these patients (Zhao et al., 2025; Kitatani et al., 2022; Woo-ko et al., 2026; Scataglini et al., 2025), the issue of economic efficiency persists. While the relevance of post-stroke and Parkinson's rehabilitation as well as the impact of RAS on this neurorehabilitation is undeniable, it is essential that further research bridges the economic and technological gap regarding Artificial Intelligence and this form of music therapy for patients.

General music therapy has been explored since the 1950s, demonstrating effects such as enhanced memory and neural integrity, reduced stress, and enhanced reward circuits while introducing non-pharmacological options in therapy (Jiao, 2025). However, research suggests that music therapy lacks application regarding research in duration and frequency of use (Liang, 2025); conversely, Rhythmic Auditory Stimulation offers a profound way of aiding patients for extended periods of time (Pacilli et al., 2025). Specifically, the ability of RAS to improve gait with stride length, gain velocity, and range of motion have positive implications for the future of

stroke recovery and quality of life improvement for Parkinson's (Woo-ko et al., 2026), stabilizing patients' gaits for periods of time ranging from five minutes to several hours (Kitatani et al., 2022). Wearable RAS also provides the opportunity for physical accessibility, applying this consistent gait stabilization (Scataglini et al., 2025; Pacilli et al., 2025). RAS is promising in the technical field as well (Zhao et al., 2025), leading to the question of how RAS and other forms of neurorehabilitation might work in conjunction with Artificial Intelligence (Liang, 2025). Due to the absence of research surrounding musical AIs in rehabilitation (Sun et al., 2024), this study analyzes the development of AI in relation to RAS. When considering AI in neurorehabilitation, it offers mental and physical accessibility, facilitating the personalization of healthcare and more enjoyable treatment (Baldisseri et al., 2022; Liang, 2025) while expanding financial accessibility. As AIs and musical AIs cost little for high output (Liang, 2025), their development alongside RAS is economically sustainable and can provide a better quality of life for Parkinson's patients and stroke recovery (Zhao et al., 2025). Therefore, the research completed in this study develops the application of Artificial Intelligence and Rhythmic Auditory Stimulation in order to bridge the gaps of AI in music-based neurorehabilitation and promotion of accessibility ranging from physical to financial.

This research models a device processing the rhythmic input of steps to determine a beats per minute (bpm) measurement with Artificial Intelligence adaptive to changing walking speeds in order to support financially accessible Rhythmic Auditory Stimulation for stroke survivors and Parkinson's disease (PD) patients. This device acts to stabilize gait in stroke recovery and PD by using input of different walking speeds collected via accelerometer to influence the output of personalized metronomic cadences via clinical asymmetry threshold (Robinson et al., 1987);

thus, effectively bridging the economic gap in the respect of these specific disorders and gait between neurorehabilitation costs and patient quality of life.

Methodology

In this research, I developed a model that tracks the pace of an individual and categorizes it into one of ten levels, outputting a metronomic cadence (bpm) in order to stabilize gait. This project is materialized in the form of two Micro:bits (Micro:bits 1A and 1B) strapped to a user's right and left ankles, respectively, as well as a second Micro:bit (Micro:bit 2) attached to the same individual's arm.

Micro:bits 1A and 1B act as detectors for input, utilizing machine learning (ML) to collect pace-related data. In order to execute these detectors, I determined the average gait velocity for stroke recovery patients as 0.4 m/s (Ponche et al., 2022) and the average gait velocity for idiopathic Parkinson's disease (IPD) as 0.94 m/s (Paker et al., 2015), as well as the upper and lower bounds for each of these groups. I then used this information to develop a range of common walking speeds for stroke recovery and IPD, ranging from 0.2 m/s to 1.1 m/s. I categorized these speeds into ten levels, using a 10% increase in walking speed for each level due to this percentage's ability to provide a gentle yet efficient transition in gait speed for patients (Bella et al., 2017). I used the formula $\text{Cadence} = \frac{v}{L} \times 60$ to convert each speed into a cadence (bpm) integer value using a baseline stride length of 0.7 m. This baseline was determined via average anthropometric gait data for impaired individuals (Paker et al., 2015). I then defined 11 variables in the Micro:bit CreateAI ML Basic Neural Network model, the first being "Still," and the next ten following the format of "Level 1," "Level 2," etcetera. In order to

train the Artificial Intelligence model, I used a metronome to collect 50 samples of walking data for each of the 10 levels, as well as the one "Still" level, by attaching Micro:bit 1 to my ankle and using the accelerometer to measure triple-axis vector-based data for $<a_x, a_y, a_z>$. This data allows the Micro:bit to detect movement and categorize it into one of the 10 levels, permitting comprehensible identification of pace.

Micro:bits 1A and 1B were programmed with Python by implementation of the radio station one, transmit power of seven, and a frequency band of 70 to facilitate transmission to Micro:bit 2 and prevent external WiFi device interference. Transmit power seven is the highest transmit power offered for Micro:bit. The function *updatePace()* uses the parameter *level* so the function can transmit the current level to Micro:bit 2 via the radio.send_number command. To increase efficiency, a selection statement implements an early return if the level parameter is equal to *current_level*, avoiding the excess time of sequencing through the rest of the function. The current level is defined as a negative one as a null value to prevent the function from returning immediately without sequencing initially, considering the user's level will begin with zero. Eleven functions (one for each speed level) call *updatePace* with a respective level, zero through 10. After each function's initialization, the CreateAI function command *ml.on_start()* iteratively evaluates the ML model for each speed level by prompting event validation. Both 1A and 1B contain the same code, although Micro:bit 1B offsets level by 100 so Micro:bit 2 can distinguish between left and right input before neutralizing the offset and averaging them.

Micro:bit 2's program functions to calculate SI, detect gait asymmetry, and output RAS using transmissions from Micro:bits 1A and 1B. Micro:bit 2 establishes the same radio group, transmit power, and frequency band as the other Micro:bits, as well as several variables. These

include *right_level*, *left_level*, *target_cadence*, *cadence_int* for the main function for RAS. The variables for ras activity and override are defined as false. Several variables for SI are also defined, including the last right and left updates, the constant *SI_TRIGGER* equal to 15 to indicate gait asymmetry, and current SI. All walking speeds and corresponding cadences are formulated into two separate arrays. Then, the function *compute_symmetry_index()* works to calculate symmetry index by using the speed array with an index of *right_level* for the variable *r* and *left_level* for the variable *l*, indicating right and left speeds, and using these values in Robinson's formula for SI ($SI = 2 \times \frac{|\bar{T}_R - \bar{T}_L|}{\bar{T}_R + \bar{T}_L} \times 100\%$) to calculate *current_si*. The function *verify_gait_regularity()* then calls the SI function to define the variable *si*, and activates RAS if *si* is greater than the constant *SI_TRIGGER* should *ras_override* be inactive. Subsequently, the function *on_received_number(receivedLVL)* uses its parameter to record the last right and left updates, call the function *verify_gait_regularity()*, calculate an average level based on right and left level, determine a target cadence using the cadence array, and to calculate *cadence_int*, representing the time interval in ms between auditory pulse intervals. This function is called every time Micro:bit 2 receives a number from either Micro:bit 1A or Micro:bit 1B. Then, the function *play_ras_loop()* plays the tone C6 for 40 ms between every cadence interval (*cadence_int*) , specifically using the variable *remainder* (*cadence _int - elapsed*) to produce a metronomic beat acting as RAS while displaying a heart icon to indicate RAS activity. This function is continuously executed in the background. The last two functions act when each button on the Micro:bit is pressed. Specifically, the function *on_button_pressed_a()*, RAS is set as active if it was previously inactive, and inactive if it was previously active. If RAS is active, the string "V" is displayed, and the string "X" is displayed if RAS is inactive. The final function

on_button_pressed_b() displays the cadence value (*target_cadence*) as a string and SI (*current_si*) as a string.

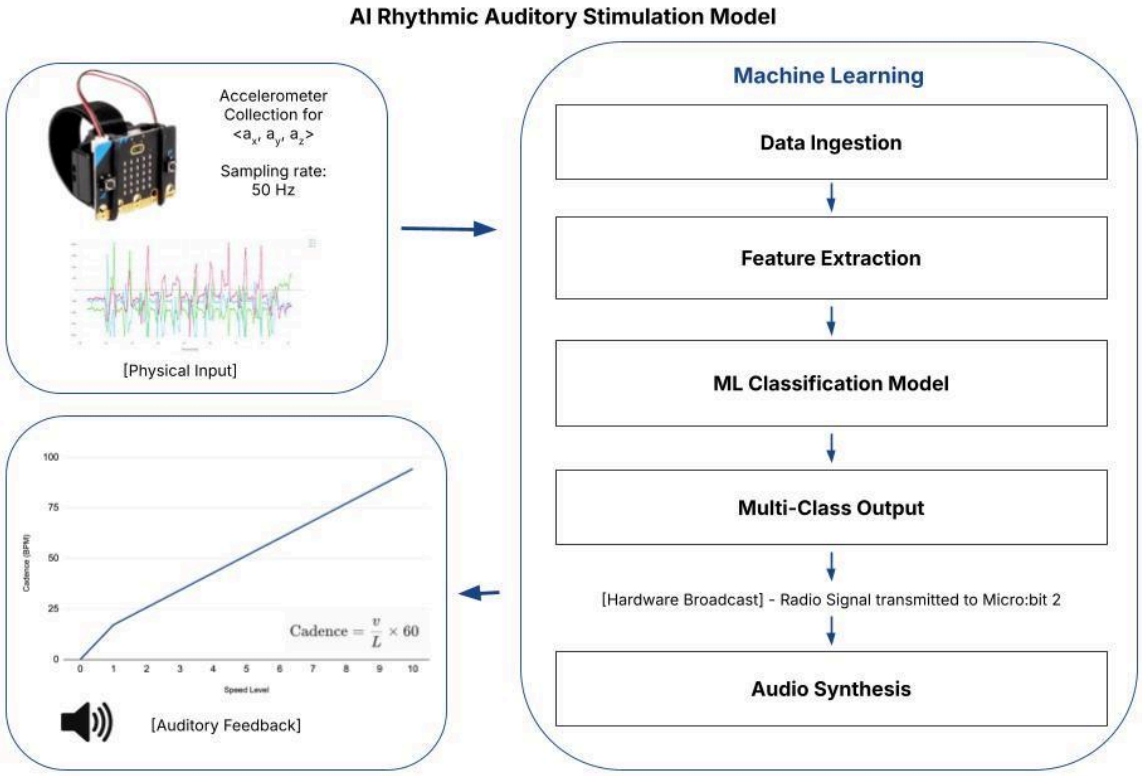


Figure 1: AI Rhythmic Auditory Stimulation Model Overview
The closed-loop system draws accelerometer data at 50 Hz from Micro:bit 1 strapped to the user's ankle. Time and frequency domain features are drawn and processed via a shallow neural network to predict one of eleven distinct walking speed levels. Predicted speed level is transmitted via radio signal to synthesize adaptive auditory feedback (BPM) based on the user's cadence.

Following implementation of the program on all three Micro:bits, data collection was conducted. In this iteration, gait asymmetry detection using SI was not verified. Data were collected for four performance metrics: (1) actual input velocity compared with predicted velocity, (2) real-time cadence adaptation to changes in walking speed, (3) node activation patterns of Micro:bits 1A and 1B, and (4) auditory pulse intervals at a target cadence of 60 bpm.In order to do so, the function *serial.write_line()* was used to display data in the device terminal. To evaluate the accuracy of the walking-speed classification algorithm, the manually

assigned input speed level was compared with the speed level calculated by the system. This was done by manually adjusting the variable *true_test_level* and comparing it with *receivedLVL*. During testing, the target input speed level (true_test_level) was maintained by walking in time with a metronome. The metronome tempo corresponded to the cadence calculated from the desired walking speed using an assumed baseline stride length of 0.7 m. Walking continued until the terminal displayed a stable value for receivedLVL. To evaluate the responsiveness of the adaptive rhythmic auditory stimulation algorithm, the calculated speed level (*computed_level*), output cadence (*target_cadence*), and running time (*input.running_time()*) were displayed in the terminal to show the relationship between velocity level and cadence, so they may be analyzed side-by-side. In order to collect this information, the participant walked continuously for 60 s while the target input speed level was increased by one approximately every 5 s by adjusting the metronome tempo. Subsequently, to verify stability of transmission between Micro:bits, data regarding node activation of Micro:bits 1A and 1B was also gathered by displaying “1” in the terminal for Micro:bit 1A activity and “0” for Micro:bit 1B activity, along with running time (*input.running_time()*). Data was gathered by walking at a steady, unmeasured pace for 15 s, allowing for node activation over time to be visualized and analyzed for system stability. Finally, auditory pulse intervals were recorded to verify the temporal consistency of RAS. The variable *last_beep_time* was displayed in the terminal and updated by subtracting the previous pulse time from each subsequent pulse event (*start_tick*). Pulse intervals were collected during 500 s of continuous walking.The pulse intervals were gathered over a period of 500s of consistent walking. All data was collected on a flat, carpeted, indoor surface. Terminal output from each trial was exported and documented for subsequent statistical analysis.

The structure of the AI Rhythmic Auditory Stimulation model implements computational, hardware, economic, and clinical efficiency. Contrary to the reliance of current RAS devices on static patient profiles, this device physically adapts to user needs. Early returns, array mapping, and the lightweight CreateAI ML model permit high functionality by reducing bandwidth and allowing rapid, accessible processing without the interference of hefty software and obstructed latency. By recalculating average gait based on asynchronous broadcasts from Micro:bits 1A and 1B, Micro:bit 2 filters out desynchronizations while offering the clinical feature of visual indication of gait instability. The Micro:bit's portable size makes it convenient and endorse routine use, with this device also functioning as a remarkably sustainable economic investment due to its low cost, being a fraction of traditional clinical neurorehabilitation models. Additionally, the device aligns with human biomechanics, factoring the natural speed adjustment of 10% into its software (Bella et al., 2017) to prevent injury and stabilize recovery through gait adaptation. This occurs while trimodal vector information is gathered from the accelerometer to apply adaptive cadence normalization. Therefore, it can be concluded that this specific approach to modeling an AI-driven Rhythmic Auditory Stimulation device is significantly efficient and has many positive implications in neurorehabilitation.

Results

After the implementation of the three Micro:bit system, data collection ensued. Gait asymmetry detection in specific using SI was not validated, as it was outside the scope of this implementation; however, it was preserved in the device program. Analysis occurring during data collection aimed to validate system performance with RAS using four primary performance

metrics, being (1) AI velocity prediction accuracy, (2) cadence adaptation, (3) rhythmic node synchronization, and (4) temporal consistency of auditory output. These analyses characterize computational accuracy, adaptive responsiveness of the system, synchronization behavior, and the temporal stability of the RAS system.

As seen in Figure 2, velocity prediction accuracy was analyzed by comparing actual input velocity with the predicted velocity of the system. Specifically, input velocity level is displayed on the horizontal axis, while predicted velocity level is displayed on the vertical axis. Each point represents one test trial of the AI system. Points in which horizontal and vertical values are equal indicate perfect predictions and verified computational accuracy, meaning that a fully linearized, diagonal line is ideal. Figure 2 portrays a nearly linear relationship between actual and predicted velocity levels, with points clustering heavily around the trendline. This indicates that increasing velocity corresponds to increasing prediction in a direct linear relationship with few deviations. The velocity prediction model demonstrated strong performance, achieving an R^2 value of 0.983. The mean absolute error was 0.182 velocity levels, while the root mean square error was 0.426 velocity levels. Exact classification accuracy was 81.8%, with no prediction differing from the true velocity by more than one velocity level. The R^2 value of 0.983 indicates high linear

agreement, strong predictive capability, and minimal estimation error.

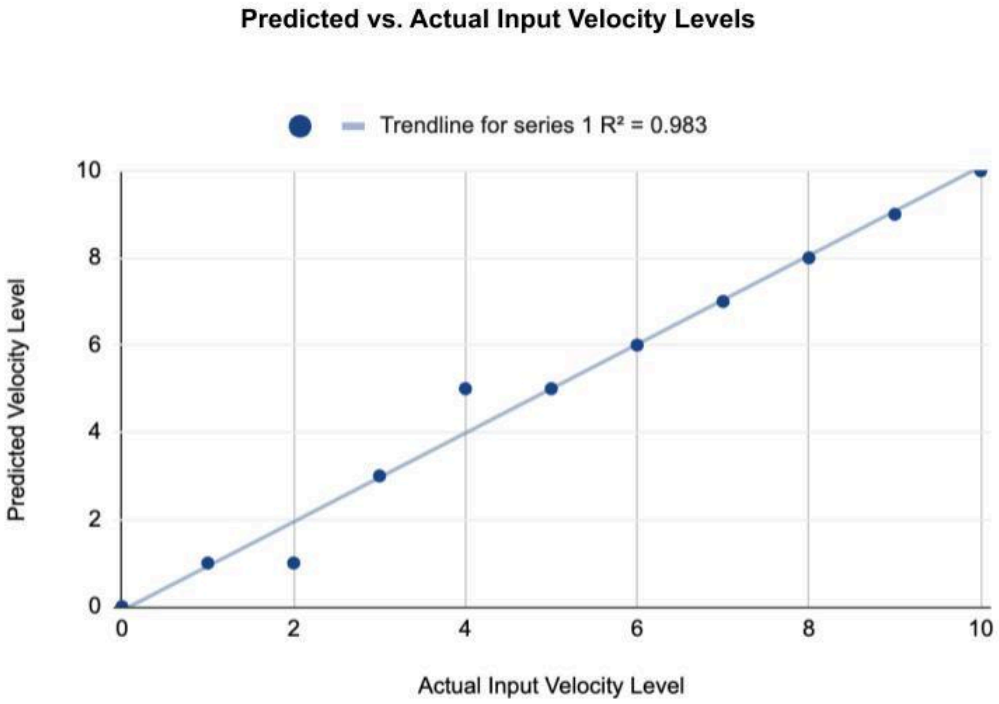


Figure 2: Evaluation of the model's velocity prediction performance with comparison to actual input levels. The trendline demonstrates a strong linear correlation ($R^2 = 0.983$), demonstrating highly accurate estimation across the tested velocity range.

As seen in Figure 2, input Velocity Level 2 has a prediction value of 1, which is an underestimate, while Input Velocity Level 4 has a prediction value of 5, indicating an overestimate. These small prediction errors are present, but do not indicate systematic bias. They are likely due to sensor variation or threshold transitions. Ultimately, Figure 2 indicates that AI can reliably classify velocity changes, which is necessary for adaptive rhythmic cueing. The accurate estimation established in this figure provides the foundation for cadence adjustment and demonstrates the device's computational accuracy in conversion from Input Velocity Level to Predicted Velocity Level.

Figure 3 portrays cadence adaptation in order to verify the adaptive responsiveness of the AI-RAS system. This figure shows velocity over time and cadence over time in a dual-axis

graph, acting to compare velocity level and cadence to indicate how quickly and accurately the system responds to a changing velocity level. Following each change in velocity level, cadence was updated after a mean latency of 1.12 ± 0.43 s (range: 0.67–1.99 s). Ten velocity transitions were evaluated across the 50.67 s trial, and cadence increased from 0 bpm at initialization to 94 bpm by the conclusion of the experiment. Although two velocity transitions did not produce an immediate cadence increase, these reflected slightly offset mapping intervals rather than missed detections. Overall, the results demonstrate consistent and responsive adaptation of rhythmic auditory stimulation to changing walking speeds. This allows the conclusion to be drawn that cadence changes after enough information is drawn regarding velocity levels, which aligns with the device program in which *target_cadence* is reliant on transmissions from Micro:bits 1A and 1B regarding level (*right_level* and *left_level*) before its value is computed.

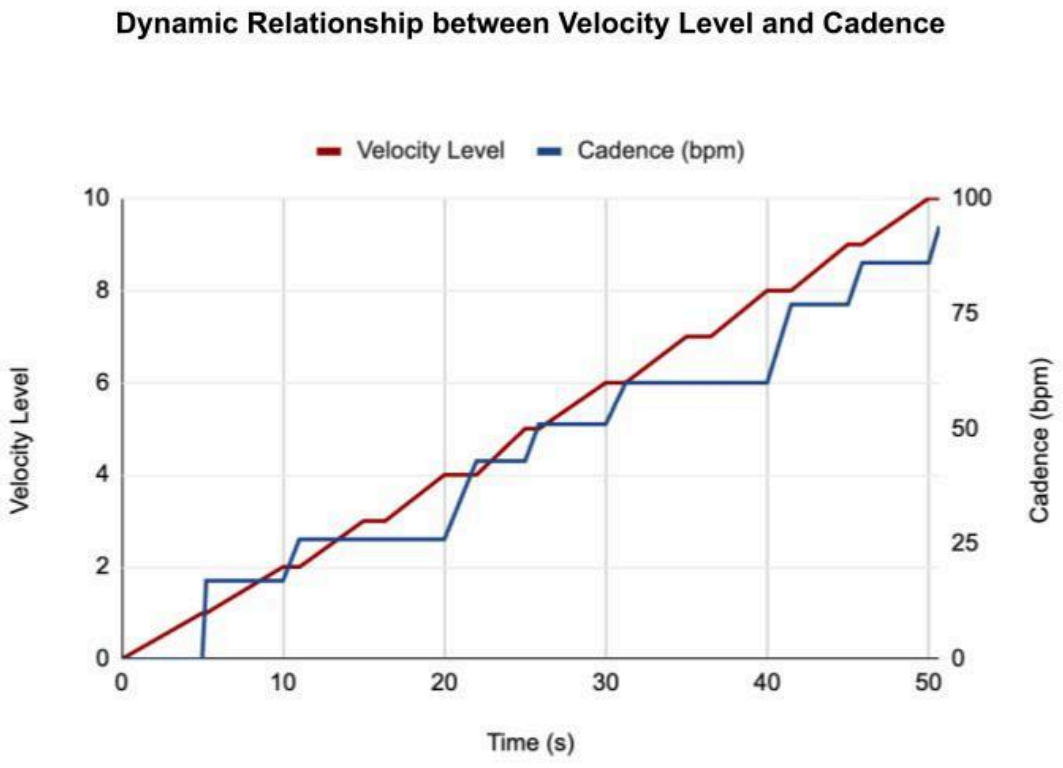


Figure 3: Response analysis showing cadence (bpm) and velocity level as a function of time. The progression displays sensor tracking latency.

Latency in cadence response can be visualized in Figure 3, indicating slight lags before aligning to be proportional with velocity level. This reflects sensor processing, averaging, and velocity classification occurring within the Micro:bit system, indicating that small latency in this sense is expected. Adaptive RAS verification requires the ability of a device to detect changes and update rhythm, which are capabilities exemplified in Figure 3. This figure indicates that the device is responsive and consistent, with the effective ability to respond and adapt to speed changes in model usage.

Figure 4 demonstrates rhythmic node synchronization, verifying the AI-RAS device's ability to synchronize transmissions from Micro:bits 1A and 1B with stability through Micro:bit 2. The blue line indicates transmissions from Micro:bit 1A (Right), while the red line indicates transmissions from Micro:bit 1B (Left). While the horizontal axis portrays time, the vertical indicates activity, with "0" indicating no activity and "1" indicating an active transmission. The graph demonstrates stable oscillations with consistent periodic behavior, regular waves, and alternating peaks. The phase difference amounts to approximately 180°, visually showing stability in alternation. Node activation occurred with a mean interval of 610.4 ± 4.8 ms, corresponding to an oscillation frequency of 1.64 Hz (approximately 98.3 activations per minute). Twenty-four activation events were recorded during the 14.04 s observation period. Activation intervals varied by only 20 ms (600–620 ms), producing a coefficient of variation of 0.79%, indicating highly stable rhythmic timing. Synchronization prevents overlap and missed pulses and is therefore crucial to the working of this device. It is clear the device demonstrates synchronization as well, as stable oscillation and consistent period length are observable in Figure 4. Twelve activations were generated by each node, yielding 100% alternation with no

simultaneous activations or missed transitions. Ultimately, this oscillatory behavior represents stable gait cue generation, confirming the stability of the system.

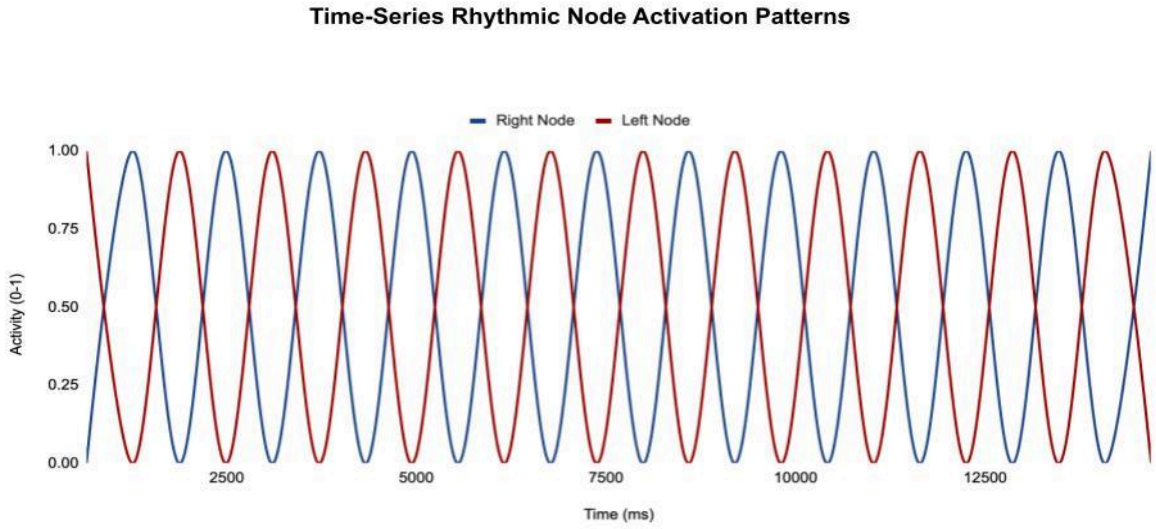


Figure 4: Continuous activation patterns of the right (blue) and left (red) controller nodes over a 15,000 ms time segment. Smooth oscillations demonstrate synchronization and rhythmic output providing control signals to Micro:bit 2.

Temporal stability of auditory output is analyzed in the histogram in Figure 5, confirming the metronomic capabilities of the RAS system. A total of 500 pulse intervals were analyzed to evaluate temporal stability of the auditory stimulation. The mean pulse interval was 999.998 ± 0.90 ms, closely matching the target interval of 1000 ms. Observed intervals ranged from 997 to 1003 ms, corresponding to a maximum deviation of 3 ms and a coefficient of variation of 0.090%. Furthermore, 89.0% of pulse intervals occurred within ±1 ms of the target interval, while 98.4% occurred within ±2 ms, demonstrating highly consistent timing performance. The spread is relatively small with no drift, indicating a consistent set of values that verify the RAS system. Precision of intervals is high, with low variability indicating consistent metronome timing. These findings are extremely significant, as stable rhythms displayed in the histogram

indicate reliable RAS. Efficient RAS leads to better synchronization with gait, facilitating the achievement of the device's primary purpose.

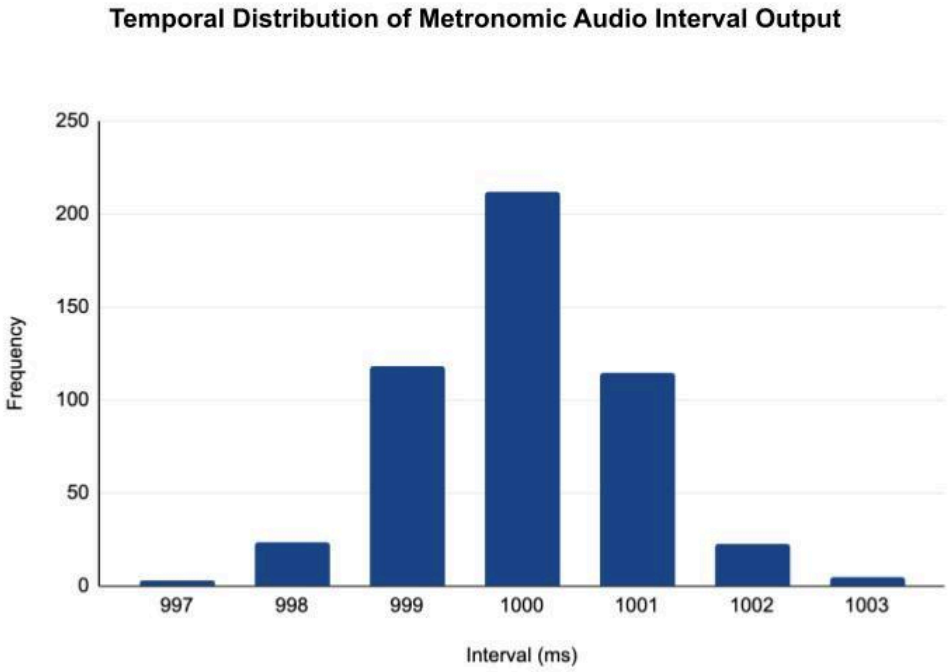


Figure 5: Distribution of 500 consecutive auditory pulse intervals generated from Micro:bit 2 at a locked cadence of 60 bpm. Intervals reflect data gathered over a 500 s walking period with a consistent speed of 0.6 m/s.

The analyses completed in this iteration of the study demonstrated high velocity prediction accuracy, successful cadence adaptation, synchronized node behavior, and temporally consistent auditory output. Collectively, these findings indicate that the proposed artificial intelligence algorithm successfully estimates walking speed and dynamically generates RAS that adapts to changing gait conditions while maintaining stable temporal accuracy.

Discussion

The results of this study indicate that the proposed AI algorithm successfully performed the primary functions required for adaptive RAS. Specifically, the system demonstrated accurate velocity prediction, responsive cadence adaptation, synchronized communication between wearable sensors, and stable auditory timing. These findings suggest that AI can effectively provide the computational framework necessary to support adaptive RAS, allowing rhythmic cues to respond automatically to changes in walking speed rather than relying on a fixed cadence.

The high velocity prediction accuracy observed in Figure 2 demonstrates that the machine learning model was capable of consistently estimating walking speed across the tested range of velocities. Accurate velocity classification is a critical component of adaptive RAS because cadence generation depends entirely on the system's ability to recognize changes in gait. Although several minor prediction errors occurred, these were limited to adjacent velocity levels and did not demonstrate a consistent tendency to overestimate or underestimate walking speed. This suggests that the classification model remained stable despite natural variability in sensor measurements. Because the prediction error remained low throughout testing, the results indicate that the algorithm can provide a reliable computational basis for adaptive rhythmic cue generation.

The cadence adaptation observed in Figure 3 further demonstrates that the system responded appropriately to changing walking speeds. Rather than maintaining a constant rhythm, the generated cadence increased as velocity increased, allowing the auditory stimulation to remain proportional to gait. A short processing delay was consistently observed following velocity transitions before cadence adjustments occurred. This delay is expected, as the system requires sufficient sensor data before computing an updated walking velocity and transmitting a

new cadence value. The consistent adaptation observed throughout the trial suggests that the algorithm successfully balanced responsiveness with computational stability, preventing frequent fluctuations while still allowing the rhythmic cues to adjust in real time.

Synchronization between the wearable Micro:bits also contributed substantially to overall system performance. Figure 4 demonstrates consistent alternation between the right and left nodes with regular activation intervals and no overlapping transmissions. Reliable synchronization is particularly important in wearable rehabilitation systems, as irregular communication between sensors can introduce timing errors that affect cadence generation. Stable oscillatory behavior observed throughout testing indicates that communication among the three Micro:bits remained consistent, supporting continuous monitoring of gait while maintaining accurate coordination between the sensing and cueing components of the device.

Similarly, the temporal stability demonstrated in Figure 5 supports the reliability of the RAS generated by the system. Rhythmic cueing is most effective when auditory pulses occur at consistent intervals, allowing users to synchronize their gait with an external rhythm. The narrow distribution of pulse intervals observed during testing indicates that the metronome maintained highly precise timing throughout data collection with minimal variability. This consistency is essential because irregular auditory timing could reduce synchronization between the user and the external rhythm. Instead, the observed stability suggests that the device is capable of generating dependable rhythmic cues suitable for adaptive gait assistance.

Collectively, findings from all four analyses demonstrate that the individual components of the proposed system function together as intended. Accurate velocity estimation enables appropriate cadence selection, synchronized communication ensures reliable transfer of

information between wearable nodes, and temporally stable auditory output provides consistent rhythmic stimulation. These results demonstrate that the components operate as an integrated system capable of adapting RAS according to changes in walking speed, representing an important step toward more intelligent RAS systems that automatically adjust to an individual's gait without requiring manual intervention or predetermined cadence settings.

While the AI-RAS device demonstrated promising computational performance during preliminary testing, several limitations should be acknowledged when interpreting these findings. For instance, validation of the model was performed using a single healthy participant, and gait asymmetry was simulated rather than collected from individuals with neurological disorders. Consequently, the present study establishes technical feasibility but does not confirm clinical effectiveness for individuals recovering from stroke or living with Parkinson's disease. Human gait varies considerably across age, height, injury severity, disease progression, and individual biomechanics; therefore, additional participant testing is necessary before generalizing these findings to broader patient populations.

Additionally, the ML component of the AI-RAS device is similarly limited by the scope of its training and evaluation. Walking speed was categorized into eleven predefined velocity levels, providing an effective framework for prototype development but limiting the model's ability to represent the continuous nature of human gait. Furthermore, the algorithm was evaluated under controlled walking conditions and was not trained to distinguish between pathological gait patterns, environmental disturbances, or unrelated movements such as foot tapping or abrupt directional changes. While the classification accuracy observed during testing demonstrates promising computational capability, additional datasets representing diverse gait characteristics would improve the robustness and generalizability of future ML models.

Hardware limitations also influenced several aspects of system performance. The Micro:bit accelerometer sampling rate restricted the temporal resolution available for gait analysis, making implementation of a clinically accurate Symmetry Index (SI) difficult within the present iteration of the device. As SI calculations require precise spatiotemporal measurements from consecutive ipsilateral steps, the one-second polling interval limits direct implementation of conventional symmetry calculations. Consequently, gait asymmetry detection relied upon differences in ML velocity classifications rather than continuous biomechanical measurements. Furthermore, the Micro:bit's accelerometer may not reliably capture subtle low-amplitude tremors commonly associated with Parkinson's disease, and the present study did not evaluate battery life, long-term device durability, wireless interference, or extended wearability.

Finally, testing was conducted within a controlled indoor environment on relatively level walking surfaces. The system was not evaluated during common real-world walking conditions such as stair negotiation, uneven terrain, obstacle avoidance, outdoor environments, or rapid changes in direction. Likewise, long-term usability, participant comfort, and rehabilitation adherence were not assessed during this preliminary investigation. These environmental and practical limitations indicate that additional validation under realistic daily living conditions will be necessary before the AI-RAS device can be considered for widespread clinical or home-based implementation.

While the first iteration of the AI-RAS device demonstrates encouraging technical feasibility, substantial opportunities remain for future development and validation. The most immediate priority is comprehensive human participant testing involving both healthy individuals and populations experiencing neurological gait impairments, including stroke

survivors and individuals with Parkinson's disease. Such testing would allow evaluation of the system under clinically relevant conditions while determining whether adaptive RAS generated by the AI-RAS device produces measurable improvements in gait symmetry, cadence, walking speed, mobility, and overall quality of life. Expanding participant diversity with respect to age, gait characteristics, and disease severity would also improve understanding of the model's generalizability across a wider range of users.

Future iterations should also focus on expanding the ML capabilities of the system beyond discrete walking speed classification. Rather than relying upon predefined intervention thresholds, adaptive algorithms could continuously learn individual walking characteristics and automatically personalize cadence outputs according to each user's rehabilitation progress. ML models capable of recognizing gradual deterioration in gait quality, predicting instability before significant asymmetry develops, or identifying early indicators of fall risk may further improve therapeutic effectiveness. Such personalization would allow rehabilitation to evolve alongside each patient's changing neurological condition rather than relying upon generalized cadence mappings.

Additional biomechanical measurements could further improve the clinical accuracy of gait analysis. Although the present study primarily utilized walking speed and bilateral velocity classifications, future implementations may incorporate stride length, step length, stance time, swing time, and double-support time to provide a more comprehensive assessment of gait quality. As stride length naturally varies according to walking velocity, future computational models incorporating adaptive stride length estimation through methods such as Taylor Series approximation or alternative mathematical approaches may improve cadence generation accuracy. Furthermore, implementation of higher-frequency accelerometer sampling, direct

heel-strike detection, or additional inertial sensors such as gyroscopes and pressure sensors could enable clinically accurate SI calculations while improving sensitivity to subtle gait abnormalities.

Hardware refinement also represents an important direction for continued development. Future prototypes should prioritize improved ergonomic design, increased battery life, enhanced wireless reliability, and greater long-term wearability to maximize user comfort during daily rehabilitation. Integration with smartphone applications or cloud-based platforms may additionally allow rehabilitation data to be stored, visualized, and shared remotely with clinicians. This connectivity would enable long-term monitoring of rehabilitation progress while supporting personalized treatment recommendations without requiring frequent in-person clinical appointments.

Finally, future research should evaluate the long-term clinical effectiveness of AI-driven RAS through longitudinal rehabilitation studies. While the present investigation establishes computational feasibility, future work should determine whether continuous use of the AI-RAS device produces sustained improvements in mobility, gait symmetry, fall risk reduction, independence, and quality of life over weeks or months of rehabilitation. Beyond stroke and Parkinson's disease, adaptive AI-RAS technologies may also prove beneficial for individuals recovering from traumatic brain injury, orthopedic surgery, multiple sclerosis, cerebral palsy, or age-related gait decline. Continued collaboration among computer scientists, rehabilitation specialists, neurologists, physical therapists, and music therapists will be essential for translating this prototype into a clinically effective rehabilitation technology capable of improving patient outcomes on a greater scale.

This study successfully developed and evaluated a wearable AI-driven RAS system capable of adapting auditory cueing according to real-time gait characteristics. By integrating ML, bilateral gait analysis, wireless communication, and wearable Micro:bit technology, the AI-RAS device accurately classified walking speed, generated adaptive cadence outputs, maintained synchronized communication between wearable components, and initiated RAS upon detection of severe simulated gait asymmetry. The strong linear relationship between walking speed and generated cadence ($R^2 = 0.983$), synchronized bilateral node behavior, and temporally consistent auditory pulse generation collectively demonstrate the computational reliability of the proposed system.

These findings support the technical practicality of combining AI and RAS within an affordable wearable platform designed for adaptive neurorehabilitation. While clinical effectiveness cannot yet be established due to preliminary testing with a healthy participant and simulated gait asymmetry, the successful implementation of the computational framework provides an encouraging proof of concept for future rehabilitation technologies. By continuously adapting therapeutic output according to changing gait characteristics, the AI-RAS device demonstrates the potential to provide personalized rehabilitation that extends beyond traditional clinical environments into everyday life.

Ultimately, this research contributes to the growing intersection of AI, wearable technology, and music-based neurorehabilitation by demonstrating that adaptive gait assistance can be implemented using inexpensive, portable hardware without sacrificing computational functionality. Continued refinement, expanded clinical validation, and interdisciplinary collaboration may allow future iterations of the AI-RAS device to improve mobility,

independence, and quality of life for individuals living with neurological gait impairments while increasing accessibility to rehabilitation worldwide.

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